

Shay Mordechai

Security Researcher | Vulnerability Discovery & Protocol Analysis

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Summary

Targeting the **XDR Research Engineer** position. Security Researcher with 3 Zero-Days (CVEs pending) via custom AI fuzzing. Expert in TCP/IP analysis, network forensics, and Zero-Trust. Advancing expertise in **Malware Analysis, Threat Hunting, and Detection Engineering** to mitigate emerging threats.

Professional Experience

DevSecOps Intern | Israel Tax Authority | Aug 2024 – Jul 2025

- Orchestrated transition to Checkmarx One, integrating automated SAST/SCA security gates into 15+ CI/CD pipelines.
- Vulnerability Triage & Detection:** Developed Python scripts to filter noise and prioritize critical threats (SQLi, DOM XSS, Supply Chain). Guided R&D remediation by translating scan results into defensive coding patterns.

Combat Soldier | Combat Intelligence Collection Corps, IDF (Hesder) | Aug 2016 – Apr 2018 - Executed covert reconnaissance and visual anomaly detection to build actionable operational target profiles.

Education

B.Sc. Computer Science | Jerusalem College of Technology | 2021-2025 | GPA: 89

Coursework: Network Analysis (97), Reverse Engineering (93), Information Security (90).

Continuous Learning: Advanced Malware Analysis (In Progress) – Mastering static/dynamic analysis, IoCs extraction, unpacking, and custom Yara rules for threat detection.

Technical Projects & Research (Details in Portfolio)

Vulnerability Research – SBCL Compiler (3 Zero-Days / CVEs) - Managed full-cycle research via Python fuzzing, uncovering 3 Zero-Days (CVEs pending). Authored technical documentation for MITRE and National CERT (#11265).

Network Protocol & Browser Security Research - Discovered DNS Rebinding attack vectors threatening local networks; proposed deterministic TTL-hardening mitigations (validated by TP-Link PSIRT). Published analysis on RWX memory allocation & CFI bypasses in V8 JIT engines.

TLS 1.2 Handshake & PCAP Simulator - Engineered a 4,000+ line **Python-based tool** simulating TLS handshakes; used **Wireshark** and **tcpdump** to dissect complex **TCP/IP** flows and generate SSLKeyLogs.

Automated Research Engine ("Predator") - Developed automated security solutions from scratch, integrating AI tools to process data and detect anomalous patterns in real-time.

LeadFlow AI (Zero-Trust SaaS Platform) | Founder & Lead Developer - Architected a secure B2B platform using **Python (FastAPI)** and Next.js. Eliminated the public attack surface (0 inbound ports) on AWS by routing traffic exclusively via **Cloudflare Tunnels** and rootless containers.

Technical Skills

- Security & Research:** XDR/SIEM Concepts, Threat Hunting, Malware Analysis, Detection Logic, Network Forensics, Zero-Day Discovery (Fuzzing), Yara Rules, IoCs.
- Networking:** Wireshark, TCP/IP, DNS, SSL/TLS, HTTP, Scapy, Zero-Trust Architecture.
- Dev & Data:** Python (FastAPI, Automation), SQL, AI/LLM integration.